

Yap Hong Ping

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Agentic AI | RAG | NLP | Computer Vision | MLOps

Applied AI Engineer | Data Scientist | 8+ Years Production AI

Full-stack AI Engineer specializing in production-grade NLP, RAG, and agentic AI systems. Built and deployed LangChain ReAct agents serving 100+ queries/day across 1,000+ enterprise documents. Deep experience in chunking strategy optimization, SFT-Mistral embedder fine-tuning, and end-to-end ML lifecycle management using Docker, Kubernetes, and GitOps.

Key Strengths

- **Data Science:** Frames ambiguous problems into measurable AI objectives; iterates with stakeholders from POC to production
- **AI Application:** Delivers ML, Deep Learning (CV and NLP), and Generative AI (LLM/RAG) solutions across domains
- **Full-Stack Deployment:** Integrates models into web applications (React.js, Vue.js, Streamlit) and APIs (FastAPI)
- **Stakeholder Engagement:** Agile delivery with regular demos, retrospectives, and cross-functional collaboration

Core Skills | Python

- **ML / Deep Learning:** PyTorch, TensorFlow, Scikit-learn, HuggingFace (BART, RoBERTa-MNLI)
- **Agentic AI and RAG:** LangChain (ReAct, tool-calling), OpenAI AsyncOpenAI, Elasticsearch kNN/HNSW, RAGAS evaluation
- **Backend and Full-Stack:** FastAPI, FastMCP, React.js (TypeScript/SSE), Vue.js, Streamlit
- **MLOps:** GitLab CI/CD, ArgoCD, Helm, Artifactory, OpenShift (dev/staging/prod)
- **AI Platforms:** Datarobot, Dataiku, Dify, SAP Joule, Copilot Studio, GitHub Copilot

PROFESSIONAL EXPERIENCE

Infineon Technologies Asia Pacific - Staff Analyst, Data Scientist (Nov 2021-Present)

Agentic Process Knowledge RAG

- Delivered multi-agent process consultant across 2 knowledge sources (Confluence, ARIS) in under 2 weeks using GitHub Copilot SpecKit for spec-driven development
- Designed Planner to 5 Phase Sub-Agents to Checker architecture with ExecutionPlan/SubGoal dependency graph; SufficiencyEvaluator triggers re-planning when coverage gaps detected
- Citation-gated answers with asyncio parallelism; React.js + TypeScript frontend with SSE streaming and per-message feedback
- **Stack:** Python, OpenAI AsyncOpenAI, FastAPI, React.js, GitLab CI/CD, ArgoCD, OpenShift

Document Retrieval and Agentic RAG Chatbot

- 3-generation evolution across 1,000+ documents (TF-IDF+SBERT to Elasticsearch kNN to LangChain ReAct agent); 96% query time reduction (2hr to 5min)
- Benchmarked 5 chunking strategies (fixed, sentence, semantic, recursive, agentic); selected agentic chunking + SFT-Mistral embedder via RAGAS evaluation

- 3 role-gated LangChain tools: abbreviation lookup, kNN semantic search, LLM-to-SQL bridge for natural-language Oracle clause queries
- **Stack:** Python, LangChain, Elasticsearch (kNN HNSW), SFT-Mistral, Docling, FastAPI

NLP Contract Clause Extraction

- Fine-tuned RoBERTa-large-MNLI (8-class logistics classifier) + BART seq2seq extraction over 200 labelled contracts; weighted F1=0.8
- Evaluated BERT, RoBERTa, CUAD, SetFit before selecting RoBERTa-MNLI; Nougat transcription + DETR table detection; integrated as LangChain tool
- **Stack:** Python, HuggingFace (RoBERTa-MNLI, BART), PyTorch, Nougat, DETR, FastAPI

Travel Expense Receipt Intelligence

- SqueezeNet CNN (7 doc classes) + Faster RCNN segmentation + EasyOCR (custom weights) + regex NER + 7-flag rule validator; 70% POC accuracy
- Benchmarked LayoutLM, Donut for hotel folios; GPT-4.1 multimodal post-project confirmed LLM-first approach for varied layouts
- **Stack:** Python, PyTorch (SqueezeNet, Faster RCNN), EasyOCR, FastAPI, SAP Concur

Distributor Invoice Audit Automation

- Configurable rule engine for AP invoice reconciliation (100+ invoices/month): 6 matching flags (SMD, currency, qty, price, SAP#, date)
- Fuzzy matching (thefuzz) + LLM for CJK company names; LLM extraction for multilingual invoices (Chinese, German)
- **Stack:** Python, ABBYY Vantage, FastAPI, Vue.js, Elasticsearch, Logstash, Docker
- **Team and Delivery:** Agile 2-week sprints, stakeholder demos and retrospectives; intern mentoring via structured onboarding

PSA Corporation - Principal Engineer, Smart Systems Solutions (Jul 2018-Nov 2021)

- **Predictive Maintenance (Maritime Equipment):** Semi-supervised pipeline (anomaly detection to clustering to classification) on time-series sensor data; reduced unplanned downtime
- **CV Equipment Monitoring (POC):** Object detection for structural crack and roller alignment inspection; deployed on edge devices
- **Cloudera Hadoop data pipelines** for large-scale industrial sensor ingestion and feature engineering

PSA Corporation - Senior Mechanical Engineer (Jul 2013-Jul 2018)

- Automated component life tracking and failure pattern analysis; translated mechanical domain knowledge into ML feature engineering

EDUCATION

- **Master of Technology (Knowledge Engineering)**, National University of Singapore - ISS (2018-2020)
- **Bachelor of Engineering (Mechatronic Engineering)**, Nanyang Technological University (2009-2013)